Bivariate Truncated Distribution

1 Bivariate Truncated Distribution

Let, $f_{X,Y}(x,y)$ be the PDF of a bivariate rv (X,Y) with sample space. $\mathcal{X} = \{(x,y) : f_{X,Y}(x,y) > 0\}$. Let $A \subset X$. Here, the pdf of (x,y) over A is,

$$g_{XY}(x,y) = \begin{cases} \frac{f_{XY}(x,y)}{P[(x,y)\in A]}, & \text{if } (x,y) \in A \\ 0 & \text{ow} \end{cases}$$

Then, $E\left(X^r Y^s \mid A\right) = \iint_A x^r y^s \frac{f_{X,Y}(x,y)}{P[(x,y)\in A]}$.

1. Let, X, Y be iid Exponential RV with mean $\lambda = 1$, define $Z = \begin{cases} 1 & X < Y \\ 0 & \text{ow} \end{cases}$. Find $E(X \mid Z = 1)$ is $U(0,1)$ and find $E(X \mid X < Y)$

$$\begin{aligned} E(X \mid Z=1) &= 2\int_0^\infty \left(\int_0^y \tfrac{x}{\lambda} e^{-x/\lambda} dx\right) \tfrac{1}{\lambda} e^{-y/\lambda} dy. && \tfrac{x}{\lambda} = z \\ &= 2\int_0^\infty \tfrac{1}{\lambda} e^{-y/\lambda} \left(\int_0^{y/\lambda} z e^{-z} dz\right) \lambda dy && dx = \lambda dz \\ &= 2\int_0^\infty e^{-y/\lambda} \left(1 - e^{-y/\lambda} - \tfrac{y}{\lambda} e^{-y/\lambda}\right) dy \\ &= 2\int_0^\infty e^{-y/\lambda} - 2\int_0^\infty e^{-\frac{2y}{\lambda}} dy - 2\int_0^\infty \tfrac{y}{\lambda} e^{-2y/\lambda} dy \\ &= 2\lambda \int_0^\infty \tfrac{1}{\lambda} e^{-y/\lambda} dy - \lambda \int_0^\infty e^{-2y/\lambda} \tfrac{2dy}{\lambda} - 2\lambda \int_0^\infty \tfrac{z}{2} \tfrac{e^{-2} dz}{2} \\ &= 2\lambda - \lambda - \tfrac{\lambda}{2} = \lambda - \tfrac{\lambda}{2} = \tfrac{\lambda}{2} \end{aligned}$$

2. Let $(X,Y) \sim \text{BN}\left(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho\right)$. Show that, $E(Y \mid X \geqslant k) = \mu_y + \rho\sigma_y \frac{\phi\left(\frac{k-\mu_x}{\sigma_x}\right)}{1-\phi\left(\frac{\mu-\mu_x}{\sigma_x}\right)}$

$$
\begin{aligned}
E(Y \mid X \geqslant k) &= \int_{-\infty}^{\infty}\int_{k}^{\infty} y\frac{f_{X,Y}(x,y)}{P(X>k)}dxdy \\
&= \int_{k}^{\infty}\left\{\int_{-\infty}^{\infty} y\frac{f_{X,Y}(x,y)}{f_X(x)}dy\right\}\frac{f_X(x)}{P(X\geqslant k)}dx \\
&= \int_{k}^{\infty} E(Y \mid X=x)\frac{f_X(x)}{P(X\geqslant k)}dx \\
E\left\{\left(\frac{Y-\mu}{\sigma_y}\right)\middle|\ X\geqslant k\right\} &= \int_{k}^{\infty} E\left(\frac{Y-\mu_y}{\sigma_y}\middle|\ X=x\right) \\
&= \int_{k}^{\infty}\rho\left(\frac{X-\mu_x}{\sigma_x}\right)\frac{\frac{1}{\sigma_x}\Phi\left(\frac{x-\mu_x}{\sigma_x}\right)}{1-\Phi\left(\frac{\mu-\mu_x}{\sigma x}\right)} \quad \frac{f_X(x)}{P[X\geqslant k]}dx \\
&= \rho\int_{\frac{k-\mu_x}{\sigma_x}}^{\infty} u\frac{\phi(u)}{1-\Phi\left(\frac{\mu-\mu_x}{\sigma_x}\right)}du, \quad \frac{x-\mu_x}{\sigma_x}=u \\
&= \frac{\rho}{1-\Phi\left(\frac{\mu-\mu_x}{\sigma_x}\right)}\int_{\frac{k-\mu_x}{\sigma_x}}^{\infty} u\phi(u)du \\
&= \frac{\rho\phi\left(\frac{k-\mu_x}{\sigma_x}\right)}{1-\Phi\left(\frac{\mu_x+\mu_x}{\sigma_x}\right)} \\
\Rightarrow E(Y|X\geq k) &= \mu_y + \rho\sigma_y\frac{\phi\left(\frac{k-\mu_x}{\sigma_x}\right)}{1-\Phi\left(\frac{k-\mu_x}{\sigma_x}\right)}
\end{aligned}
$$

3. If $(X,Y) \sim \text{BN}(0,0,1,1,\rho)$, show that, $\rho_{X^2Y^2} = \rho^2$
$X \sim N(0,1), \quad Y \sim N(0,1)$.
Also, $\forall X = x, Y|X \sim N\left(\rho x, 1-\rho^2\right)$
$\rho_{X^2Y^2} = \frac{E\left(X^2Y^2\right)-E\left(X^2\right)E\left(Y^2\right)}{\sqrt{E(X^4)-E^2(X^2)}\sqrt{E(Y^4)-E^2(Y^2)}}$

$$\therefore E\left(X^2\right) = V(X) + E^2(X) = 1. \quad E\left(X^4\right) = 3\sigma^4 = 3$$

Similarly $E\left(Y^2\right) = 1, \quad E\left(Y^4\right) = 3.$

$$
\begin{aligned}
E\left(X^2Y^2\right) &= E\left[E\left(X^2Y^2 \mid X\right)\right]. \\
&= E\left[X^2E\left(Y^2 \mid X\right)\right] = E\left[X^2\left\{V(Y \mid X) + E^2(Y \mid X)\right\}\right] \\
&= E\left\{X^2\left(1-\rho^2+\rho^2X^2\right)\right\} = \left(1-\rho^2\right)\cdot 1 + \rho^2\cdot 3 = 1+2\rho^2
\end{aligned}
$$

$$\rho_{X^2Y^2} = \frac{1+2\rho^2-1}{\sqrt{3-1}\sqrt{3-1}} = \rho^2$$

4. If $(X,Y) \sim BN(0,0,1,1,\rho)$, find the dist of $U = \max(X,Y)$.
$F_U(u) = P[U \leqslant u] = P[\max(X,Y) \leqslant u] = P[X \leqslant u, Y < u]$.

$$= \int_{-\infty}^{u}\int_{-\infty}^{u} f_{X,Y}(x,y)dxdy = \int_{-\infty}^{u} g(u,y)dy \quad (say)$$

PDF of U is $f_U(u) = \frac{d}{du}F_U(u)$. [Leibnitz Rule of differentiability on integral]

$$\therefore f_Y(u) = \frac{\partial}{\partial u}\int_{-\infty}^{u} g(u,y)dy. \quad g(u,y) = \int_{-\infty}^{u} f_{X,Y}(x,y)dx$$

$$= \int_{-\infty}^{u} f_{x,y}(u,y)dy + g(u,u)\cdot 1 - 0$$

$$= \int_{-\infty}^{u} f_{X,Y}(u,y)dy + \int_{-\infty}^{u} f_{X,Y}(x,y)dx$$

$$= \int_{-\infty}^{u} \{f_{X,Y}((u,y) + f_{X,Y}(y,u)\}\, dy$$

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(x^2+y^2-2p\right)} \text{ on simplification.}$$

$$f_U(u) = 2\phi(u)\Phi(u\frac{1+\rho}{1-\rho})]$$

5. $\binom{X_1}{X_2} \backsim N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right)$

Define, $Y_1 = X_1 \operatorname{sign}(X_2), Y_2 = X_2 \operatorname{sign}(X_1)$, where sign $(x) = \begin{cases} -1 & x < 0 \\ +1 & x \geqslant 0 \end{cases}$

a) Find the marginal distribution of Y_1 and Y_2.
b) Show that $(Y_1 Y_2)$ does not follow BVN.
c) Find $\rho(Y_1, Y_2)$.

$$\begin{aligned} &(X_1, X_2) \sim \text{BN}(0,0,1,1,0) \\ \Rightarrow &X_1 \sim N(0,1) \rightarrow \text{ independent or } \rho = 0 \\ &X_2 \sim N(0,1) \end{aligned}$$

a) $F_{Y_1}(y_1) = P[Y_1 \leqslant y_1]$

$$= P[Y_1 \leqslant y_1; X_2 < 0] + P[Y_1 \leqslant y_1; X_2 > 0]$$
$$= P[-X_1 \leqslant y_1; X_2 < 0] + P[X_1 \leqslant y_1; X_2 > 0]$$
$$= P[-(-X_1) \leqslant y_1, X_2 < 0] + P[X_1 \leqslant y_1; X_2 > 0] \quad \left(\text{ as } X_1 \overset{D}{=} -X_1 \sim N(0,1)\right)$$
$$= P[X_1 \leqslant y_1] \quad [P(A \cap B) + P(A \cap B^C) = P(A)]$$
$$= F_{X_1}(y_1) \text{ Hence } Y_1 \overset{D}{=} X_1 \sim N(0,1).$$

Similarly, $Y_2 \sim N(0,1)$.

b) $Y_1 Y_2 = X_1 \operatorname{sign}(X_1)\, X_2 \operatorname{sign}(X_2) = |X_1|\,|X_2| \geqslant 0$ wp 1.
Hence, $Y_1 Y_2$ cannot take -ve values.

$$\Rightarrow (Y_1, Y_2) \text{ does not follow BVN}$$